Problem 3

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Problem 1 *Suppose that* $\int_1^3 f(x)\,dx = 4$

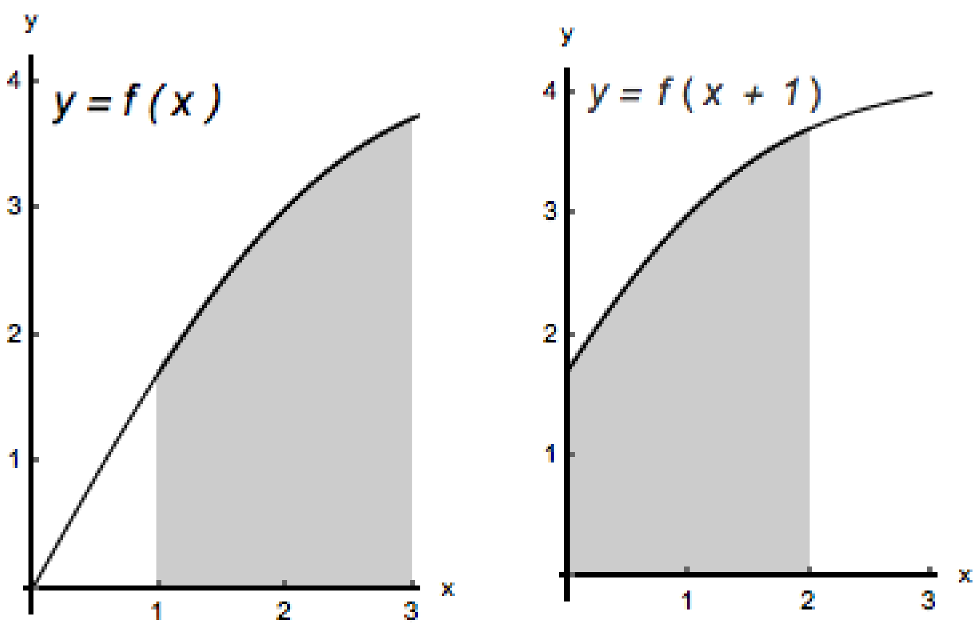
(a) *Evaluate the following integrals.*

(i) $\int_0^2 f(x+1)\,dx$

Explanation. Let $u = x + 1$. Then $du = dx$; when $x = 0$, $u = 1$; when $x = 2$, $u = 3$. Therefore,

$$\int_0^2 f(x+1)\,dx = \int_1^3 f(u)\,du = 4$$

Notice that the region whose net area is given by the integral $\int_0^2 f(x+1)\,dx$ is just the region whose net area is given by the integral $\int_1^3 f(x)\,dx$, only "shifted to the left by 1". Obviously, their net areas have to be equal. We illustrate this in the figure below.



(ii) $\displaystyle\int_1^9 3\frac{f(\sqrt{x})}{\sqrt{x}}\,dx$

Explanation. Let $u=\sqrt{x}$. Then $du=\dfrac{1}{2\sqrt{x}}\,dx$; when $x=1$, $u=1$; when $x=9$, $u=3$. Therefore,

$$\int_1^9 3\frac{f(\sqrt{x})}{\sqrt{x}}\,dx=\int_1^9 3\cdot 2\cdot\frac{f(\sqrt{x})}{2\cdot\sqrt{x}}\,dx=\int_1^3 6f(u)\,du=6\int_1^3 f(u)\,du=6\cdot 4=24$$

(iii) $\displaystyle\int_{\frac{1}{3}}^1 f(3x)\,dx$

Explanation. Let $u=3x$. Then $du=3\,dx$; when $x=\dfrac{1}{3}$, $u=1$; when $x=1$, $u=3$. Therefore,

$$\int_{\frac{1}{3}}^1 f(3x)\,dx=\int_{\frac{1}{3}}^1\frac{3}{3}f(3x)\,dx=\int_1^3\frac{1}{3}f(u)\,du=\frac{1}{3}\int_1^3 f(u)\,du=\frac{1}{3}\cdot 4=\frac{4}{3}$$

(iv) $\displaystyle\int_2^4 3f(x-1)\,dx$

Explanation. Let $u=x-1$. Then $du=dx$; when $x=1$, $u=1$; when $x=4$, $u=3$. Therefore,

$$\int_2^4 3f(x-1)\,dx=\int_1^3 3f(u)\,du=3\int_1^3 f(u)\,du=3\cdot 4=12$$

(v) $\displaystyle\int_0^{\sqrt{2}} 3xf(x^2+1)\,dx$

Explanation. Let $u=x^2+1$. Then $du=2x\,dx$; when $x=0$, $u=1$; when $x=\sqrt{2}$, $u=3$. Therefore,

$$\int_0^{\sqrt{2}} 3xf(x^2+1)\,dx=\int_0^{\sqrt{2}} 3\cdot\frac{2}{2}\cdot xf(x^2+1)\,dx=\int_1^3\frac{3}{2}\cdot f(u)\,du=\frac{3}{2}\cdot\int_1^3 f(u)\,du=\frac{3}{2}\cdot 4=6$$

(b) *Assume that f is odd. Evaluate* $\displaystyle\int_{-3}^{-1} f(x)\,dx$

Explanation. Since f is odd, then $f(-x)=-f(x)$ or $f(x)=-f(-x)$, for all x in its domain.
Let $u=-x$. Then $du=-\,dx$; when $x=-3$, $u=3$; when $x=-1$, $u=1$.

Therefore,

$$\int_{-3}^{-1} f(x)\,dx = \int_{-3}^{-1} [-f(-x)]\,dx = \int_{3}^{1} f(u)\,du = -\int_{1}^{3} f(u)\,du = -4$$

(c) *Assume that f is even. Evaluate* $\displaystyle\int_{-3}^{-1} f(x)\,dx$

Explanation. Since f is even, then $f(-x) = f(x)$, for all x in its domain. Let $u = -x$. Then $du = -\,dx$; when $x = -3$, $u = 3$; when $x = -1$, $u = 1$.

Therefore,

$$\int_{-3}^{-1} f(x)\,dx = \int_{-3}^{-1} [f(-x)]\,dx = -\int_{-3}^{-1} [f(-x)](-\,dx) = -\int_{3}^{1} f(u)\,du = \int_{1}^{3} f(u)\,du = 4$$